

# Divisibility and the Division Algorithm

- **Definition 1**

Given  $a, b \in \mathbb{Z}$  we say that  $a$  **divides**  $b$ , denoted  $a \mid b$ , if there exists  $k \in \mathbb{Z}$  such that  $b = ka$ .

We say that  $a$  is a **factor** or **divisor** of  $b$ .

We say that  $b$  is **divisible** by  $a$  or  $b$  is a **multiple** of  $a$ .

If  $a$  does not divide  $b$  we write  $a \nmid b$ .

The set of all divisors of  $b$  is denoted  $\text{Div}(b)$  and the set of all positive divisors of  $b$  is denoted  $\text{Div}_+(b)$ .

- **Remark 1**

Since  $0 = k \cdot 0$  for all  $k \in \mathbb{Z}$  we have

$$0 \mid b \iff b = 0.$$

Note that  $a \mid 0$  for all  $a \in \mathbb{Z}$  since  $0 = 0 \cdot a$  for all  $a \in \mathbb{Z}$ . Besides,  $1 \mid a$  for all  $a \in \mathbb{Z}$  since  $a = a \cdot 1$  for all  $a \in \mathbb{Z}$ .

If  $a \neq 0$  and  $a \mid b$  the integer  $k$  such that  $b = ka$  is unique. In fact, if  $a \neq 0$  and  $ka = b = k'a$  then by the cancellation law  $k = k'$ .

The unique  $k$  such that  $b = ka$  is called the **exact quotient** of  $b$  by  $a$  and is denoted  $\frac{b}{a}$ .

- **Example 1.1**

$7 \nmid 6$  because there is no  $k \in \mathbb{Z}$  such that  $6 = 7k$ .

$6 \mid 42$  since  $42 = 7 \cdot 6$  and we have that  $7 = \frac{42}{6}$ .

$\text{Div}(-14) = \{-1, 1, -2, 2, -7, 7, -14, 14\}$ .

$\text{Div}_+(-14) = \{1, 2, 7, 14\}$ .

- **Proposition 1.2**

Let  $a, b \in \mathbb{Z}$ . If  $a \mid b$  then  $\text{Div}(a) \subseteq \text{Div}(b)$  and  $\text{Div}_+(a) \subseteq \text{Div}_+(b)$ .

- **Proof 1.2.1**

Suppose that  $a \mid b$  then by definition there exists  $k \in \mathbb{Z}$  such that  $b = ka$ . Let  $q \in \text{Div}(a)$  then  $q \mid a$ , that is, there exists  $l \in \mathbb{Z}$  such that  $a = lq$ . Hence

$$b = ka = k(lq) = (kl)q \quad \text{with } kl \in \mathbb{Z}.$$

Then  $q \mid b$ , that is,  $q \in \text{Div}(b)$  and thus  $\text{Div}(a) \subseteq \text{Div}(b)$ .

Analogously, we have that  $\text{Div}_+(a) \subseteq \text{Div}_+(b)$ .

- **Theorem 1.3 (Properties of divisibility)**

Let  $a, b, c \in \mathbb{Z}$ .

1.  $a \mid b \iff a \mid -b \iff -a \mid b \iff -a \mid -b$ .
2. If  $a \mid b$  then  $a \mid bc$ .
3. If  $a \mid b$  then  $ac \mid bc$ .
4. If  $a \mid b$  and  $a \mid c$  then  $a \mid \alpha b + \beta c$  for all  $\alpha, \beta \in \mathbb{Z}$ .
5. If  $a \mid (b + c)$  and  $a \mid b$  then  $a \mid c$ .
6. If  $a \mid b$  and  $b \neq 0$  then  $0 < |a| \leq |b|$ .
7. Divisibility is an order relation on  $\mathbb{N}$ .

- **Proof 1.3.1**

1. ( $\implies$ ) If  $a \mid b$  then by definition there exists  $k \in \mathbb{Z}$  such that  $b = ka$ , hence  $-b = -(ka) = (-k)a$  with  $-k \in \mathbb{Z}$  and thus  $a \mid -b$ .

( $\Leftarrow$ ) If  $a \mid -b$  then there exists  $k \in \mathbb{Z}$  such that  $-b = ka$ , hence  $b = -(ka) = (-k)a$  with  $-k \in \mathbb{Z}$  and thus  $a \mid b$ .

The remaining proofs are similar.

2. If  $a \mid b$  then there exists  $k \in \mathbb{Z}$  such that  $b = ka$ , hence  $bc = (ka)c = (kc)a$  with  $kc \in \mathbb{Z}$  and thus  $a \mid bc$ .
3. If  $a \mid b$  then there exists  $k \in \mathbb{Z}$  such that  $b = ka$ , hence  $bc = (ka)c = k(ac)$  and thus  $ac \mid bc$ .
4. If  $a \mid b$  and  $a \mid c$  then there exist  $k, k' \in \mathbb{Z}$  such that  $b = ka$  and  $c = k'a$ . Hence if  $\alpha, \beta \in \mathbb{Z}$  then

$$\begin{aligned}\alpha b + \beta c &= \alpha(ka) + \beta(k'a) \\ &= (\alpha k + \beta k')a\end{aligned}$$

with  $\alpha k + \beta k' \in \mathbb{Z}$  and thus  $a \mid \alpha b + \beta c$ .

5. If  $a \mid (b + c)$  and  $a \mid b$  then by Property 4 we have that  $a \mid (b + c) - b = c$ .
6. If  $a \mid b$  then there exists  $k \in \mathbb{Z}$  such that  $b = ka$  and since  $b \neq 0$  we have that both  $k$  and  $a$  are nonzero. Hence  $|b| = |ka| = |k||a| \geq 1 \cdot |a| = |a| > 0$ .
7. **Reflexive** Let  $a \in \mathbb{N}$ .

Since  $a = 1 \cdot a$  and  $1 \in \mathbb{Z}$ , by the definition of divisibility we have  $a \mid a$ , which means  $a \sim a$ .

Therefore, the relation is reflexive.

**Antisymmetric** Let  $a, b \in \mathbb{N}$  such that  $a \sim b$  and  $b \sim a$ .

By definition,  $a \mid b \implies b = k_1 a$  for some  $k_1 \in \mathbb{Z}$ , and  $b \mid a \implies a = k_2 b$  for some  $k_2 \in \mathbb{Z}$ .

Substituting  $a = k_2 b$  into the first equation gives:

$$b = k_1(k_2 b) = (k_1 k_2)b.$$

Since  $b \in \mathbb{N}$ , we can divide both sides by  $b$  to get  $k_1 k_2 = 1$ . Since  $a, b \in \mathbb{N}$ , their multipliers  $k_1, k_2$  must be positive integers. The only positive integers satisfying  $k_1 k_2 = 1$  are  $k_1 = k_2 = 1$ .

Substituting  $k_1 = 1$  back yields  $b = 1 \cdot a = a$ .

Therefore, the relation is antisymmetric.

**Transitive** Let  $a, b, c \in \mathbb{N}$  such that  $a \sim b$  and  $b \sim c$ .

By definition,  $a \mid b \implies b = k_1 a$  for some  $k_1 \in \mathbb{Z}$ , and  $b \mid c \implies c = k_2 b$  for some  $k_2 \in \mathbb{Z}$ .

Substituting the expression for  $b$  into the equation for  $c$  yields:

$$c = k_2(k_1 a) = (k_2 k_1)a.$$

Since  $k_1, k_2 \in \mathbb{Z}$ , their product  $k_2 k_1 \in \mathbb{Z}$ . Thus, by the definition of divisibility, we have  $a \mid c$ , which means  $a \sim c$ .

Therefore, the relation is transitive.

Since the divisibility relation on  $\mathbb{N}$  is reflexive, antisymmetric, and transitive, it is an order relation.

#### ◦ Question 1.4

If  $a \mid bc$ , must  $a \mid b$  or  $a \mid c$ ?

##### ■ Solution 1.4.1

No. For example,  $6 \mid (2)(3)$  but  $6 \nmid 2$  and  $6 \nmid 3$ .

Divisibility is not an order relation on  $\mathbb{Z}$  because it is not antisymmetric. For example,  $2 \mid -2$  and  $-2 \mid 2$  but  $2 \neq -2$ .

$\neq -2$ .

- **Definition 2**

Given  $n, d \in \mathbb{Z}$  with  $d \neq 0$ , two integers  $q, r$  are called the **quotient** and the **remainder** of  $n$  by  $d$  if

$$n = dq + r \quad \text{with } 0 \leq r < |d|.$$

- **Example 2.1**

The signs of  $n$  and  $d$  affect the quotients but the remainder is always nonnegative. For example:

$$\begin{aligned} 23 &= 5 \cdot 4 + 3 \\ -23 &= 5 \cdot (-5) + 2 \\ 23 &= (-5) \cdot (-4) + 3 \\ -23 &= (-5) \cdot 5 + 2 \end{aligned}$$

- **Proposition 2.2**

Given  $n, d \in \mathbb{Z}$  with  $d \neq 0$ , the quotient and the remainder of the division of  $n$  by  $d$  are unique.

- **Proof 2.2.1**

Suppose that

$$dq + r = n = dq' + r' \quad \text{with } 0 \leq r, r' < |d|.$$

Then we obtain that  $d(q' - q) = r - r'$  and thus  $d \mid r - r'$ . Since  $0 \leq r, r' < |d|$ , we have  $|r - r'| < |d|$ . By Property 6, a nonzero integer cannot be divided by an integer with a larger absolute value, so this is possible if and only if  $r - r' = 0$ , that is,  $r = r'$ . Now,  $dq = dq'$  with  $d \neq 0$ , then by the cancellation law  $q = q'$ .

- **Theorem 2.3 (The division algorithm)**

Given  $n, d \in \mathbb{Z}$  with  $d \neq 0$ , there exist unique  $q, r \in \mathbb{Z}$  such that

$$n = dq + r \quad \text{with } 0 \leq r < |d|.$$

- **Proof 2.3.1**

Let  $m = |d|$  and consider the set  $S = \{n - km : k \in \mathbb{Z} \wedge n - km \geq 0\}$ .

Note that  $S \neq \emptyset$  since if  $n \geq 0$ , then

$$n - 0 \cdot m = n \geq 0,$$

that is,  $n \in S$ , and if  $n < 0$ , then

$$n - n \cdot m = n \cdot (1 - m) \geq 0 \quad \text{since } m \geq 1 \text{ and } 1 - m \leq 0,$$

that is,  $n - nm \in S$ .

Now, as  $S \subseteq \mathbb{Z}$  with  $S \neq \emptyset$  and bounded below by 0, then by the well-ordering principle for integers,  $S$  has a first element  $r = n - km$  for some  $k \in \mathbb{Z}$ . If  $r \geq m$ , then

$$r - m = n - km - m = n - (k + 1)m \geq 0.$$

So  $r - m \in S$ , which contradicts the minimality of  $r$ , therefore  $0 \leq r < m = |d|$ . If  $d > 0$ , then  $n = dq + r$  with  $q = k$ , and if  $d < 0$ , then  $n = dq + r$  with  $q = -k$ .

- **Corollary 2.4**

Every integer is either even or odd, but not both.

- **Proof 2.4.1**

Let  $n \in \mathbb{Z}$  and apply the division algorithm with divisor  $d = 2$ . Then there exist unique  $q, r \in \mathbb{Z}$  such that

$$n = 2q + r \quad \text{with } 0 \leq r < |2| = 2.$$

Since  $r \in \mathbb{Z}$  and  $0 \leq r < 2$ , the remainder  $r$  must be either 0 or 1. We consider these two cases for existence and uniqueness:

**1. Existence:**

- If  $r = 0$ , then  $n = 2q$ , which means  $n$  is an even number by definition.
- If  $r = 1$ , then  $n = 2q + 1$ , which means  $n$  is an odd number by definition.

Therefore, every integer is either even or odd.

**2. Uniqueness:**

Suppose for the sake of contradiction that an integer  $n$  is both even and odd.

By definition, there exist  $k_1, k_2 \in \mathbb{Z}$  such that  $n = 2k_1$  and  $n = 2k_2 + 1$ .

This implies we have found two different representations for the division of  $n$  by 2:

$$n = 2k_1 + 0 \quad \text{and} \quad n = 2k_2 + 1.$$

However, the division algorithm guarantees that the quotient and the remainder are uniquely determined. Since  $0 \neq 1$ , this contradicts the uniqueness of the remainder  $r$ .

Therefore, every integer is either even or odd, but not both.